Assignment Classification

1. Machine Learning
   Supervised Learning
   Classification
2. Total no. of Rows: 399
   Total no. of Columns: 25
3. Nominal Method
   One Hot Encoding
4. The Final Model is Logistic Regression
5. Logistic Grid Classification

```
The f1_macro value for best parameter {'penalty': 'l2', 'solver': 'newton-cg'}: 0.9924946382275899
```

```
print(cm)
```

```
[[51  0]
 [ 1 81]]
```

```
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.98      1.00      0.99        51
           1       1.00      0.99      0.99        82

    accuracy                           0.99       133
   macro avg       0.99      0.99      0.99       133
weighted avg       0.99      0.99      0.99       133
```

```
from sklearn.metrics import roc_auc_score
roc_auc_score(y_test,grid.predict_proba(X_test)[:,1])
```

SVM Grid Classification

```
The f1_macro value for best parameter {'C': 10, 'gamma': 'auto', 'kernel': 'sigmoid'}: 0.9924946382275899
```

```
print(cm)
```

```
[[51  0]
 [ 1 81]]
```

```
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.98      1.00      0.99        51
           1       1.00      0.99      0.99        82

    accuracy                           0.99       133
   macro avg       0.99      0.99      0.99       133
weighted avg       0.99      0.99      0.99       133
```

```
from sklearn.metrics import roc_auc_score
roc_auc_score(y_test,grid.predict_proba(X_test)[:,1])
```

```
1.0
```

Random Forest

```
The f1_macro value for best parameter {'criterion': 'gini', 'max_features': 'log2', 'n_estimators': 100}: 1.0
```

```
print(cm)
```

```
[[51  0]
 [ 0 82]]
```

```
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       1.00      1.00      1.00        51
           1       1.00      1.00      1.00        82

    accuracy                           1.00       133
   macro avg       1.00      1.00      1.00       133
weighted avg       1.00      1.00      1.00       133
```

```
from sklearn.metrics import roc_auc_score
roc_auc_score(y_test,grid.predict_proba(X_test)[:,1])
```

```
1.0
```

Decision Tree

```
The f1_macro value for best parameter {'criterion': 'gini', 'max_features': 'sqrt', 'splitter': 'random'}: 0.9700283472213296
```

```
print(cm)
```

```
[[50  1]
 [ 3 79]]
```

```
print(clf_report)
```

```
              precision    recall  f1-score   support

           0       0.94      0.98      0.96        51
           1       0.99      0.96      0.98        82

    accuracy                           0.97       133
   macro avg       0.97      0.97      0.97       133
weighted avg       0.97      0.97      0.97       133
```

```
from sklearn.metrics import roc_auc_score
roc_auc_score(y_test,grid.predict_proba(X_test)[:,1])
```

```
0.9719033955045432
```

6.Final model is Logistic Regression

Logistic Regression is preferred because it is a simpler, highly interpretable model that outputs well-calibrated probabilities, making it more reliable for future predictions.